

Runwei Zhou

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Overview

I am a Ph.D. in Electrical and Computing Engineering, specialized in

- AI/ML for electromagnetic (EM) and integrated circuit (IC) design and modeling
- Fast integral equation (IE)-based solvers in computational electromagnetics
- Electronic and quantum design automation
- Large scale parallel algorithms

Education

Ph.D. in ECE	Purdue University, USA	2022 - 2026
M.S. in ECE	Purdue University, USA	2022 - 2024
B.S. in Physics (<i>Honor School of Gifted Young</i>)	University of Science and Technology of China, PR China	2018 - 2022

Skills

Programming: Python (PyTorch, Scikit-learn, NumPy, Pandas), C/C++, MATLAB, LabView, Bash

Engineering Software: Cadence Spectre/Allegro, COMSOL, Solidworks, AutoCAD, Origin

Working Language: English (Fluent), Chinese (Native), Japanese (Elementary)

Professional Experience

Cadence Design System, CA, United States

Aug 2026-

Lead Software Engineer

- R&D engineer, who is involved in developing and maintaining high-efficiency cutting-edge algorithms and software for the Spectre circuit simulator.

Cadence Design System, CA, United States

May 2025-Aug 2025

Software Graduate Intern

- Developed new algorithms and software for the Spectre circuit simulator
- Implemented novel features for circuit model order reduction with performing comprehensive software validation and testing, and investigating performance/accuracy issues in the simulator.

Research Experience

On-Chip Electromagnetics Lab, Purdue University, IN, United States

Advisor: Dan Jiao

Research Assistant

Explicit Rank-Preserving Solver for Time-harmonic Maxwell's Equations with Broadband Stability

- Addressed the low-frequency breakdown problem in full-wave electromagnetic circuit analysis with a novel explicit rank-stabilizer
- Fast $O(N)$ algorithm with minor computational effort
- Successfully applied to solve large multi-scale integrated circuit problems and antenna analysis

Robust Physics-Informed Data-Free Machine Learning Method for 3-D Electromagnetic Analysis

- Proposed an unsupervised physics-informed neural network scheme for solving Maxwell's equations.
- Formulated a single objective loss function for efficient learning by integrating IE operators and FEM operators with complicated boundary conditions.

- Successfully applied it to solve large-scale IC simulation and full-wave EM scattering problems.

Error-Controlled Rank-Revealing Method for Solving Large Global Optimization Problems

- Developed a cutting-edge error-controlled rank-revealing based optimizer, surpassing many well-known optimization methods (BO, SA, GA, etc...) in both runtime efficiency and capability of identifying the global optimum.
- Successfully applied to challenging benchmark optimization problems and real-world design, including antenna optimization and floor planning in heterogeneously integrated semiconductor system design.
- Successfully applied to solving nonlinear optimization problems in machine learning.

Fast Volume-Integral-Equation-Based Modeling and Analysis of Spin Qubit Decoherence with Non-reciprocal Magnetic Metasurfaces

- Developed a rapid solver for decoherence analysis problems in quantum spin system.
- Extensive numerical experiments and validation show the capability of the proposed solver in real-world non-reciprocal anisotropic magnetic metasurfaces-controlled decoherence modeling.

Fast Well-Conditioned VIE Solver for Analyzing Nonlocal Optical Responses in Quantum Nanostructures

- Developed a novel fast solver of electromagnetic scattering problems for quantum device design.
- Accelerated the volume integral equation solver accounting for nonlocal microscopic optical response.
- Applied the solver to simulate limits of two-spin-qubit quantum gate fidelity from thermal and vacuum fluctuations.

Helmholtz-Institut Mainz, Mainz University, Rheinland-Pfalz, Germany

Advisor: Dmitry Budker

Visiting Scholar

Reduced Contrast of Nitrogen-Vacancy Ensembles in Diamond at Zero Magnetic Field

- Formulated and simulated dynamics of nuclear spins in nitrogen-vacancy centers in diamonds known for promising quantum device candidates.
- Built experimental platform and software. Achieved reasonable results with theoretical analysis.

CAS Key Laboratory of Microscale Magnetic Resonance, USTC, Anhui, China

Advisor: Xing Rong

Research Assistant

High Sensitivity Quantum Magnetometer Based on Nitrogen-Vacancy Centers in Diamond

- Designed a new schematic for compact high-sensitivity quantum magnetometry based on nitrogen-vacancy ensemble diamonds. Built prototype product. Reached a sensitivity of $500 \text{ pT/Hz}^{1/2}$, which is twice better than previous work.

Honors and Awards

Finalist Paper Award 2026 IEEE Antennas and Propagation Symposium, July 2026

Honorable Mention Paper Award 2025 IEEE Antennas and Propagation Symposium, July 2025

Outstanding Student, Silver Prize (Top 10%), Oct. 2019/20/21/22

National Competition of Mathematics, First Prize, Dec. 2019

Second prize, National Physics Olympiad, Dec 2018

Publications

[1] R. Zhou, W. Sun, S. Bharadwaj, Z. Jacob, and D. Jiao, "Fast Volume Integral Equation Based Modeling and Simulation of Nonlocal Effects in Quantum Computing Devices," IEEE International Symposium on Antennas and Propagation (AP-S), July 2023.

[2] R. Zhou and D. Jiao, "Fast Volume Integral Equation Based Modeling of Quantum Gate Circuitry," 2023 IEEE MTT S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO), June, 2023.

- [3] W. Sun, S. Bharadwaj, R. Zhou, D. Jiao, and Z. Jacob, "Computational Electromagnetics Meets Spin Qubits: Controlling Noise Effects in Quantum Sensing and Computing." IEEE Journal on Multiscale and Multiphysics Computational Techniques (2024).
- [4] R. Zhou and D. Jiao, "Efficient Neural-Network Based Solution of Integral Equations for Electromagnetic Analysis." 2024 IEEE International Symposium on Antennas and Propagation (AP-S), July 2024.
- [5] R. Zhou and D. Jiao, "Fast Well-Conditioned Volume Integral Equation Solver for Analyzing Nonlocal Optical Responses in Quantum Nanostructures." IEEE Journal on Multiscale and Multiphysics Computational Techniques (2025).
- [6] R. Zhou and D. Jiao, "Error Analysis and Error-Controlled Fast Rank-Revealing Method for Solving Global Optimization Problems." 2025 IEEE International Symposium on Antennas and Propagation (AP-S), July 2025.
- [7] R. Zhou, W. Sun, Z. Jacob, and D. Jiao, "Fast Volume-Integral-Equation-Based Modeling and Analysis of Spin Qubit Decoherence with Non-reciprocal Magnetic Metasurfaces." 2025 IEEE International Symposium on Antennas and Propagation (AP-S), July 2025.
- [8] R. Zhou and D. Jiao, "Robust Physics-Informed Data-Free Machine Learning Method for 3-D Electromagnetic Analysis" IEEE Transactions on Microwave Theory and Techniques (2026)